KING FAHAD CENTRAL HOSPITAL

MRI ACUTE ISCHEMIC STROKE IMAGING PROTOCOL

Aligned with the 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke

Stroke Center Accreditation / Certification Survey Readiness Document

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1. Purpose and Scope

This protocol defines the indications, technique, timing benchmarks, and reporting standards for brain MRI in the evaluation of suspected acute ischemic stroke (AIS). It is designed to satisfy the imaging-related requirements referenced by The Joint Commission (TJC), DNV Healthcare, and Healthcare Facilities Accreditation Program (HFAP) stroke certification programs (Primary, Thrombectomy-Capable, and Comprehensive Stroke Center levels), and to operationalize the imaging recommendations in the 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke, which replaces the 2018 guideline and its 2019 update.

This protocol applies whenever MRI is selected as the initial or adjunct modality for a Code Stroke activation, and whenever MRI is used for extended-window thrombolysis selection, wake-up stroke, stroke-mimic evaluation, or post-treatment assessment. It is intended to be used alongside — not in place of — the institution's non-contrast CT stroke protocol, which remains an acceptable Class I first-line modality under the guideline.

2. Governing Guidelines and Standards

- 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke (Stroke; supersedes the 2018 guideline and 2019 focused update).
- ACR–ASNR–ASFNR–SPR Practice Parameter for the Performance of Neuroimaging in Stroke.
- ACR Appropriateness Criteria: Cerebrovascular Disease / Suspected Stroke.
- The Joint Commission Comprehensive Stroke Center (CSC) / Primary Stroke Center (PSC) / Thrombectomy-Capable Stroke Center (TSC) certification standards, and DNV/HFAP equivalents.
- American Heart Association Get With The Guidelines–Stroke (GWTG-Stroke) performance measure definitions, used for the timing benchmarks in Section 4.

Where this protocol and a subsequent AHA/ASA update conflict, the most current published AHA/ASA guideline governs; this document must be reviewed and revised within 90 days of any new guideline release.

3. Key Guideline Recommendations Operationalized by This Protocol

COR / LOE	Recommendation	Protocol Element
1, A	Emergent brain imaging (NCCT or MRI) is recommended on initial evaluation to assess ischemic burden and exclude hemorrhage before reperfusion therapy.	Section 5 — Abbreviated Code Stroke sequence set
1, B-NR	Imaging should be performed as rapidly as possible (benchmark: initiation within 25 minutes of arrival) via process-improvement protocols.	Section 4 — Timing benchmarks
1, A	Vascular imaging (CTA/MRA) should not be delayed to obtain serum creatinine in suspected LVO.	Section 7 — Vascular imaging / safety screening
2a, B-R	For wake-up stroke or unknown onset >4.5 h from last known well, DWI-FLAIR mismatch selection can be useful for extended-window IV thrombolysis eligibility.	Section 6 — Extended-window mismatch protocol
2a, B-R	In patients 4.5–9 h from last known well (or wake-up within 9 h of sleep midpoint) with salvageable penumbra on automated perfusion imaging, IV thrombolysis may be reasonable.	Section 6 — Perfusion-based mismatch protocol

1, A	In suspected LVO within 24 h of last known well, emergent brain and vascular imaging (CT/CTA or MRI/MRA) should be performed as rapidly as possible for EVT selection.	Section 5 & 7 — MRA neck/circle of Willis
2a, B-R	Pediatric suspected AIS: emergent brain and vascular MRI/MRA is reasonable to identify LVO and differentiate stroke mimics, minimizing radiation.	Section 9 — Pediatric considerations

4. Time-Based Performance Benchmarks (GWTG-Stroke / AHA-Aligned)

These interval targets apply regardless of whether CT or MRI is the initial modality and must be captured in the stroke registry for accreditation data abstraction.

Interval	Target	Compliance Goal
Door to physician / stroke team activation	≤ 10–15 min	≥ 80%
Door to imaging initiation (CT or MRI)	≤ 25 min	≥ 50%
Door to image interpretation available	≤ 45 min	≥ 50%
Door to IV thrombolysis (needle) — primary goal	≤ 60 min	≥ 80%
Door to IV thrombolysis (needle) — stretch goal	≤ 45 min	≥ 50%
Door to groin puncture (EVT-eligible)	≤ 90 min	Institution-defined
Door to stroke unit / ICU admission	≤ 3 h	Institution-defined

Note: MRI is frequently rate-limiting versus CT. Sites using MRI as the first-line Code Stroke modality must demonstrate through PI data that the 25-minute door-to-imaging-initiation benchmark is met; if not consistently achievable, NCCT should remain the default first study with MRI reserved for mismatch/extended-window selection, stroke-mimic clarification, or posterior fossa evaluation.

5. Core MRI Sequence Protocol

5.1 Abbreviated "Code Stroke" MRI (target scan time < 10 minutes)

Used for hyperacute triage when MRI is the first-line modality or when confirming/excluding stroke mimics before thrombolysis decision-making.

Sequence	Plane	Approx. Time	Slice Thk.	Purpose
DWI (b=0, 1000) + ADC map	Axial	~1 min	4–5 mm	Detect cytotoxic edema within minutes of onset; core infarct localization
Axial GRE or SWI	Axial	~1 min	4–5 mm	Exclude hemorrhage (required before thrombolysis; substitutes for NCCT hemorrhage exclusion)
Axial FLAIR	Axial	~1.5–2 min	4–5 mm	DWI-FLAIR mismatch for onset-time estimation; chronic disease burden
3D TOF MRA, circle of Willis ± neck	Axial source	~3–4 min	1 mm (recon)	LVO detection for EVT triage (Class I within 24 h of LKW)

5.2 Comprehensive Stroke MRI (target scan time 15–20 minutes)

Adds sequences below to the abbreviated set once the patient is stable and extended-window or comprehensive-center evaluation is indicated.

Sequence	Plane	Approx. Time	Slice Thk.	Purpose
Axial T2 or T2/PD	Axial	~1.5 min	4–5 mm	Correlate lesion age; identify mimics (tumor, demyelination)
Dynamic Susceptibility Contrast (DSC) or ASL perfusion	Axial	~2–4 min	5 mm	Core/penumbra (Tmax, CBF) mismatch for 4.5–9 h / wake-up thrombolysis and 6–24 h EVT selection
Contrast-enhanced MRA neck + intracranial (if gadolinium given)	Coronal	~1–2 min	1–1.5 mm	Extracranial carotid/vertebral disease, dissection
Post-contrast T1 (optional)	Axial	~1.5 min	4–5 mm	Problem-solving: mimic exclusion, blood-brain-barrier breakdown

Automated perfusion post-processing (e.g., RAPID, Viz.ai, iSchemaView, or equivalent FDA-cleared software) should be available with results returned to the treating team within 10 minutes of acquisition completion, per comprehensive/thrombectomy-capable center certification requirements for standardized core/penumbra volumetrics.

6. Extended-Window and Wake-Up Stroke Selection Imaging

6.1 DWI-FLAIR Mismatch (unknown onset or wake-up, otherwise eligible for IV thrombolysis)

- Positive mismatch = visible DWI hyperintensity with no or only subtle FLAIR signal change in the corresponding region, indicating infarct age likely < 4.5 hours.
- Both sequences must be reviewed on the scanner console or PACS in real time by the interpreting radiologist/stroke neurologist — do not wait for final dictated report.
- Document mismatch status explicitly in the preliminary read: "DWI-FLAIR mismatch present/absent."

6.2 Perfusion-Based Mismatch (4.5–9 h from last known well, or wake-up within 9 h of sleep midpoint)

- Core infarct estimated by DWI lesion volume or ADC/CBF thresholds on automated software.
- Penumbra (Tmax > 6 s) versus core mismatch ratio and absolute mismatch volume reported per validated software output (trial-consistent thresholds, e.g., mismatch ratio ≥ 1.8 and volume ≥ 15 mL, core < institutionally defined ceiling).
- Results must be transmitted to the stroke team electronically (mobile app or PACS push) given the time-sensitive nature of the treatment decision.

7. Vascular Imaging and MRI Safety Screening

7.1 Vascular Imaging

For any patient with suspected LVO presenting within 24 hours of last known well, MRA of the cervical and intracranial vessels (TOF ± contrast-enhanced) is obtained emergently and must not be delayed pending renal function results. If MRI vascular imaging is non-diagnostic or unavailable urgently, CTA is obtained without delaying the imaging-to-decision interval.

7.2 Rapid MRI Safety Screening (must not delay the 25-minute door-to-imaging benchmark)

- Abbreviated verbal/written Code Stroke MRI screening form: implanted cardiac devices, aneurysm clips, cochlear implants, other ferromagnetic implants, pregnancy status, prior orbital metal exposure, body habitus/weight limit.
- MRI-conditional pacemakers/ICDs may proceed per device-specific conditional protocol with MRI safety officer or trained staff sign-off; non-conditional devices default to CT pathway.
- Gadolinium-based contrast is not required for the core hyperacute protocol; when used, screen for prior contrast reaction and eGFR per ACR Manual on Contrast Media, but do not delay non-contrast DWI/FLAIR/GRE/MRA acquisition or thrombolysis decision-making while awaiting eGFR.
- Patients who cannot be rapidly screened or who are agitated/uncooperative are triaged to NCCT/CTA to avoid delay.

8. Structured Reporting Requirements

The preliminary interpretation communicated to the stroke team (verbal or electronic, target ≤ 45 minutes from door) must explicitly address each of the following elements:

- Presence/absence of acute infarct on DWI, with territory and estimated volume.
- Presence/absence of hemorrhage on GRE/SWI.
- DWI-FLAIR mismatch status (present / absent / indeterminate) when onset is unknown or wake-up.
- Large vessel occlusion: vessel, segment, and side, on MRA.
- Perfusion core and penumbra volumes and mismatch ratio, when perfusion is performed, with software name/version documented.
- ASPECTS or equivalent ischemic burden score when clinically applicable and comparable to companion CT scoring used by the program.
- Alternative diagnosis / stroke-mimic findings if infarct is absent.

A standardized structured-reporting template (drop-down/macro based) is required by most accreditation surveyors as evidence of process reliability; template fields should mirror the bullet list above verbatim.

9. Pediatric Considerations

Pediatric patients (<18 years) with suspected acute stroke are outside the scope of the KFCH Adult Stroke Program. Such patients shall be managed in accordance with the hospital Pediatric Emergency, Referral, and Transfer Policy and referred or transferred to an appropriate pediatric stroke-capable facility as clinically indicated. Adult IV thrombolysis and mechanical thrombectomy eligibility criteria contained within the KFCH Adult Stroke Program shall not be automatically applied to pediatric patients. Urgent neuroimaging, when clinically required before transfer, shall be determined collaboratively by the responsible pediatric/emergency physician, radiologist/neuroradiologist, and relevant specialist, without delaying stabilization or appropriate transfer.

10. Roles and Responsibilities

Role	Responsibility
ED Physician / Stroke Team	Activate Code Stroke, complete rapid MRI safety screen, communicate last-known-well time and NIHSS to imaging.
MRI Technologist	Prioritize Code Stroke over routine schedule, execute abbreviated protocol without deviation, notify radiologist immediately upon completion.
Interpreting Radiologist / Neuroradiologist	Real-time console/PACS review, verbal preliminary read within benchmark window, complete structured report.
Stroke Neurologist / Vascular Neurologist	Integrate imaging with clinical exam for thrombolysis/EVT decision; document decision rationale referencing imaging findings.
MRI Safety Officer	Maintain conditional-device library, approve conditional scanning, review near-miss safety events.
Stroke Program Coordinator	Abstract timing/outcome data into GWTG-Stroke registry, track PI metrics in Section 4 for accreditation reporting.

11. Quality Assurance and Accreditation Readiness

- Monthly PI review of door-to-imaging and door-to-interpretation times, stratified by CT-first vs. MRI-first pathway.
- Quarterly case review of any DWI-FLAIR or perfusion-mismatch-selected thrombolysis case for outcome correlation.
- Annual competency validation for MRI technologists on the abbreviated Code Stroke protocol.
- Documented process for protocol deviation review (e.g., imaging delay beyond benchmark) with corrective action tracking — a standard survey focus area.
- Maintain this protocol, PI data, and structured-report samples in the accreditation evidence binder for on-site or virtual survey.

12. References

1. 2026 Guideline for the Early Management of Patients With Acute Ischemic Stroke: A Guideline From the American Heart Association/American Stroke Association. Stroke. (Replaces the 2018 Guideline and 2019 Update.)

2. Guidelines for the Early Management of Patients With Acute Ischemic Stroke: 2019 Update to the 2018 Guidelines. Stroke. 2019;50: e344–e418.

3. ACR–ASNR–ASFNR–SPR Practice Parameter for the Performance of Neuroimaging in Stroke.

4. ACR Appropriateness Criteria: Cerebrovascular Disease.

5. American Heart Association Get With The Guidelines–Stroke Program: performance measure specifications.

6. The Joint Commission Comprehensive / Primary / Thrombectomy-Capable Stroke Center Certification Requirements.

13. Review and Approval

This protocol requires multidisciplinary sign-off prior to activation and inclusion in the accreditation evidence file, consistent with survey expectations for a formally governed clinical protocol.

Role	Name / Signature	Date
Stroke Program Medical Director		
Neuroradiology Section Chief		
MRI / Radiology Operations Director		
Emergency Department Medical Director		
Stroke Program Coordinator		
Chief Medical Officer / Designee		

Revision History

Version	Date	Summary of Change	Approved By
1.0	____________	Initial release, aligned to 2026 AHA/ASA guideline	____________